



GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III

UNIT 1: MATRICES

LECTURE 3

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING



MATRICES LECTURE 3

TOPICS TO BE COVERED:

- Orthogonal matrix
- Complex Matrix
- Hermitian and Skew-Hermitian matrix
- Unitary matrix



Orthogonal matrix

Definition: A square matrix A is called an **Orthogonal Matrix** if $A \cdot A' = I = A' \cdot A$.

Example 1: The matrix $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ is an Orthogonal Matrix

$$A' = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\text{Since } A \cdot A' = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & \cos \theta \sin \theta - \sin \theta \cos \theta \\ \sin \theta \cos \theta - \cos \theta \sin \theta & \cos^2 \theta + \sin^2 \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A' \cdot A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & -\cos \theta \sin \theta + \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \cos \theta \sin \theta & \cos^2 \theta + \sin^2 \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since $A \cdot A' = I = A' \cdot A$, A is an orthogonal matrix.



Orthogonal matrix

Example 2: Verify that $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ is an Orthogonal Matrix.

Solution:

$$A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \quad \therefore A' = \frac{1}{3} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$$
$$AA' = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$
$$A'A = \frac{1}{9} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$\therefore A$ is an orthogonal matrix



Complex Matrix:

Complex matrix: If some or all elements of a matrix are complex number, then the matrix is called a complex matrix.

Ex. $A = \begin{bmatrix} 2 + 3i & -5 \\ i & 6 - 5i \end{bmatrix}$

Every complex matrix $A = P + iQ$ where P and Q are real matrices

$$A = \begin{bmatrix} 2 & -5 \\ 0 & 6 \end{bmatrix} + i \begin{bmatrix} 3 & 0 \\ 1 & -5 \end{bmatrix}$$

Conjugate of a matrix: The conjugate $\bar{A}$ of a complex matrix A is obtained by replacing all elements of A by their corresponding complex conjugate.

If $A = [a_{ij}]_{m \times n}$ then $\bar{A} = [\bar{a}_{ij}]_{m \times n}$

$$\bar{A} = \begin{bmatrix} 2 - 3i & -5 \\ -i & 6 + 5i \end{bmatrix}$$

Properties of conjugate: Let A and B be two complex matrices conformable for the indicated operations. Then

1. $\overline{(\bar{A})} = A$
2. $\overline{(A + B)} = \bar{A} + \bar{B}$
3. $\overline{(kA)} = \bar{k} \bar{A}$
4. $\overline{(A \cdot B)} = \bar{A} \cdot \bar{B}$



Complex Matrix:

Conjugate transpose of a matrix: The conjugate transpose A^θ of a complex matrix A is the transpose of conjugate of A

$$\text{If } A = [a_{ij}]_{m \times n} \text{ then } A^\theta = (\overline{A})' = (\bar{A})' = [\overline{a_{ji}}]_{n \times m}$$

$$\text{If } A = \begin{bmatrix} -2 + 4i & 3 - 5i & i \\ -6i & -6 - 4i & 5 + i \end{bmatrix}$$

$$\text{Then } A^\theta = \begin{bmatrix} -2 - 4i & 6i \\ 3 + 5i & -6 + 4i \\ -i & 5 - i \end{bmatrix}$$

Properties of conjugate transpose: Let A and B be two complex matrices conformable for the indicated operations. Then

1. $(A^\theta)^\theta = A$
2. $(A + B)^\theta = A^\theta + B^\theta$
3. $(kA)^\theta = \bar{k} A^\theta$
4. $(A \cdot B)^\theta = B^\theta \cdot A^\theta$



Hermitian Matrix:

Definition: A complex square matrix $A = [a_{ij}]_{n \times n}$ is said to be Hermitian if $A^\theta = A$

$$\text{i.e. } a_{ij} = \overline{a_{ji}} \quad \forall i, j$$

For diagonal elements, $i = j$

$$a_{ii} = \overline{a_{ii}}$$

$$\Rightarrow a_{ii} \text{ is real } \forall i$$

The diagonal elements of a Hermitian matrix are real numbers.

Example: Let $A = \begin{bmatrix} 1 & 2-i & 4+3i \\ 2+i & 4 & -1 \\ 4-3i & -1 & 5 \end{bmatrix}$

$$A^\theta = (\overline{A})' = \begin{bmatrix} 1 & 2-i & 4+3i \\ 2+i & 4 & -1 \\ 4-3i & -1 & 5 \end{bmatrix} = A$$

Therefore A is Hermitian matrix.



Skew-Hermitian Matrix:

Definition: A complex square matrix $A = [a_{ij}]_{n \times n}$ is said to be Skew - Hermitian if $A^\theta = -A$

$$\text{i.e. } a_{ij} = -\overline{a_{ji}} \quad \forall i, j$$

For diagonal elements, $i = j$

$$a_{ii} = -\overline{a_{ii}}$$

$\Rightarrow a_{ii}$ is zero or purely imaginary

The diagonal elements of a Skew-Hermitian matrix are either zeros or purely imaginary.

Example: Let $A = \begin{bmatrix} 4i & 2-i & 4+3i \\ -2-i & 0 & -i \\ -4+3i & -i & -5i \end{bmatrix}$

$$A^\theta = (\overline{A})' = \begin{bmatrix} -4i & -2+i & -4-3i \\ 2+i & 0 & i \\ 4-3i & i & 5i \end{bmatrix} = - \begin{bmatrix} 4i & 2-i & 4+3i \\ -2-i & 0 & -i \\ -4+3i & -i & -5i \end{bmatrix} = -A$$

Therefore A is Skew-Hermitian matrix.



Important Theorem:

Theorem : Every complex square matrix A can be uniquely expressed as sum of Hermitian and a skew-Hermitian matrix.

Proof: Let A be any square complex matrix ,

$$A = \frac{1}{2} [A + A^\theta] + \frac{1}{2} [A - A^\theta]$$
$$= P + Q \quad \text{where } P = \frac{1}{2} [A + A^\theta] \text{ and } Q = \frac{1}{2} [A - A^\theta]$$

To prove P is Hermitian and Q is skew Hermitian

$$\text{Consider } P^\theta = \left[\frac{1}{2} (A + A^\theta) \right]^\theta = \frac{1}{2} (A + A^\theta)^\theta = \frac{1}{2} [A^\theta + (A^\theta)^\theta] = \frac{1}{2} [A^\theta + A] = \frac{1}{2} [A + A^\theta] = P$$

Therefore P is Hermitian.

$$\text{Now consider } Q^\theta = \left[\frac{1}{2} (A - A^\theta) \right]^\theta = \frac{1}{2} (A - A^\theta)^\theta = \frac{1}{2} [A^\theta - (A^\theta)^\theta] = \frac{1}{2} [A^\theta - A] = - \frac{1}{2} [A - A^\theta] = -Q$$

Therefore Q is Skew- Hermitian.



Important Theorem:

Uniqueness : Let us assume that $A = R + S$, where R is Hermitian and S is skew Hermitian
Then,

$$A^\theta = (R + S)^\theta = R^\theta + S^\theta = R - S$$

$$A + A^\theta = R + S + R - S = 2R$$

$$A - A^\theta = R + S - (R - S) = 2S$$

Solving the above two equations $A + A^\theta = 2R$ and $A - A^\theta = 2S$

$$\text{we get } R = \frac{1}{2} [A + A^\theta] = P$$

$$\text{and } S = \frac{1}{2} [A - A^\theta] = Q$$

Therefore P and Q are unique.



Solved Example

1. Express the matrix $A = \begin{bmatrix} 4i & 2-3i & 5+6i \\ -3 & 4-2i & i \\ 6-i & 1+i & 0 \end{bmatrix}$ as a sum of a Hermitian and skew Hermitian matrix.

Solution: $A^\theta = \begin{bmatrix} -4i & -3 & 6+i \\ 2+3i & 4+2i & 1-i \\ 5-6i & -i & 0 \end{bmatrix}$

$$A = P + Q$$

$$\text{where } P = \frac{1}{2} [A + A^\theta] = \frac{1}{2} \left\{ \begin{bmatrix} 4i & 2-3i & 5+6i \\ -3 & 4-2i & i \\ 6-i & 1+i & 0 \end{bmatrix} + \begin{bmatrix} -4i & -3 & 6+i \\ 2+3i & 4+2i & 1-i \\ 5-6i & -i & 0 \end{bmatrix} \right\} = \begin{bmatrix} 0 & \frac{-1-3i}{2} & \frac{11+7i}{2} \\ \frac{-1+3i}{2} & 4 & \frac{1}{2} \\ \frac{11-7i}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$\text{and } Q = \frac{1}{2} [A - A^\theta] = \frac{1}{2} \left\{ \begin{bmatrix} 4i & 2-3i & 5+6i \\ -3 & 4-2i & i \\ 6-i & 1+i & 0 \end{bmatrix} - \begin{bmatrix} -4i & -3 & 6+i \\ 2+3i & 4+2i & 1-i \\ 5-6i & -i & 0 \end{bmatrix} \right\} = \begin{bmatrix} -4i & \frac{5-3i}{2} & \frac{-1+5i}{2} \\ \frac{-5-3i}{2} & 2i & \frac{-1+2i}{2} \\ \frac{1+5i}{2} & \frac{1+2i}{2} & 0 \end{bmatrix}$$



Unitary Matrix

Unitary matrix: A matrix is said to be if $A \cdot A^\theta = I = A^\theta \cdot A$

$$\text{Ex. } A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix} \text{ and } A^\theta = \begin{bmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ \frac{-1-i}{2} & \frac{1+i}{2} \end{bmatrix} \text{ and}$$

$$A \cdot A^\theta = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix} \begin{bmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ \frac{-1-i}{2} & \frac{1+i}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = A^\theta \cdot A$$

Therefore, A is unitary matrix

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Exercise

- If $A = \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$ prove that $A^{-1} = A'$,
where A' is the transpose of A .
- If $A \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$ where A is 2×2 matrix then find A .
- For a square matrix A of order n , show that $|adj A| = |A|^{n-1}$
- If A is a Hermitian matrix show that iA is skew Hermitian.

Solution

- $A \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$

- Inverse of $\begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} -1 & -1 \\ -2 & 0 \end{bmatrix}$

- $A \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \times -\frac{1}{2} \begin{bmatrix} -1 & -1 \\ -2 & 0 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -2 & 0 \end{bmatrix}$

$$A = -\frac{1}{2} \begin{bmatrix} -4 & -2 \\ 1 & 1 \end{bmatrix}$$

Solution

- To prove that $|\text{adj}A| = |A|^{n-1}$
- $A.\text{adj}A = \text{adj}A.A = |A|I$

$$|A.\text{adj}A| = ||A|I| = |A|^n$$

$$|A| |\text{adj}A| = |A|^n$$

$$|\text{adj}A| = |A|^{n-1} \text{ since } |A| \neq 0$$

$A^\theta = A$ since A is Hermetian.

$(iA)^\theta = -i A^\theta = -iA = -(iA)$ therefore iA is skew Hermetian.